Real Exchange Rate App: Submission

Oct 26, 2026 · @Obi

Links

- **Streamlit app:** real-exchange-rate-app-dbyxqkresptdzta3pz4ksd.streamlit.app
- **GitHub repository:** github.com/asinugobi/real-exchange-rate-app

The app runs `app.py` from the `main` branch on Python 3.12, with Secrets left empty. All three files (`app.py`, `requirements.txt`, `README.md`) sit at the top level of the repository.

LLM used

Model	Provider	Interface	Dates used
Claude Opus 5.5 (`claude-opus-5-5`)	Anthropic	Claude desktop app, Code tab (Claude Code)	25 and 26 October 2026

One model did all of the work: it wrote the three files, tested them against the live IMF API, created the GitHub repository, deployed to Streamlit and wrote this document.

Prompts

These are the exact prompts sent to the model, in order. Screenshots of the same chat are in the next section.

#	Date	Prompt (verbatim)
1	25 Oct 2026	"Complete the assignment below for me", followed by the full assignment text pasted in (steps 1–5, the app and economic requirements, and the data hints).
2	26 Oct 2026	"Yes, open GitHub in the in-app browser and help me create the repo and deploy to Streamlit. I'll sign in myself."

#	Date	Prompt (verbatim)
3	26 Oct 2026	"I'm signed in to GitHub. Go ahead and create the repo."
4	26 Oct 2026	"I'm signed in to Streamlit Community Cloud. Go ahead and deploy the app."
5	26 Oct 2026	"push this code to github. then pull the repo to my local machine. clone the repo to following directory /Users/obiasinugo/Code/"
6	26 Oct 2026	"Yes, copy Submission.md and the CSV into ~/Code/real-exchange-rate-app but don't commit them."
7	26 Oct 2026	"push the new changes"
8	26 Oct 2026	"I've run gh auth login. Push the commit now."
9	26 Oct 2026	"follow the submission instructions for me and create a document that i can submit."

The requirements the model built to, from the pasted assignment text:

> Allow users to choose Country 1 and Country 2 from IMF countries and the Euro area. Retrieve monthly IMF CPI and exchange-rate data from January 1976 where available. Define S(t) as units of Country 2 currency per one unit of Country 1 currency; let P1(t) and P2(t) be the respective monthly CPIs. Calculate RER(t) = S(t) × P1(t) / P2(t). Over the user-selected start and end months, calculate RER* = average of RER(t) over valid months; plot Real Index(t) = 100 × RER(t) / RER*. Recalculate RER* if dates change. Plot nominal S(t) in actual currency units—do not normalize it. Let users select Real, Nominal, or Both in interactive charts (use separate axes for Both) and provide a CSV download button of all observations and calculations within the selected sample period.

The pasted data hints gave the API base `https://api.imf.org/external/sdmx/3.0`, CPI key `{ISO3}.CPI._T.IX.M`, exchange-rate key `{ISO3}.XDC_USD.PA_RT.M`, and G163 for the Euro area.

Screenshots

The three screenshots below come from the live Streamlit app on 26 Oct 2026, with UK as Country 1, U.S. as Country 2, and the full sample (Jan 1976–Sept 2026).

Real view. Real Index with the 100 line; RER* = 1.4249 and the mean index is 100.00.

Share

Real Exchange Rate App

Monthly IMF CPI and exchange-rate data. S(t) = units of Country 2 currency per 1 unit of Country 1 currency; RER(t) = S(t) × P1(t) / P2(t); Real Index(t) = 100 × RER(t) / RER*.

Country 1 (base currency)

United Kingdom (GBR)

Country 2 (quote currency)

United States (USA)

Sample period (RER* is recalculated over these months)

1976-01 — 2026-09

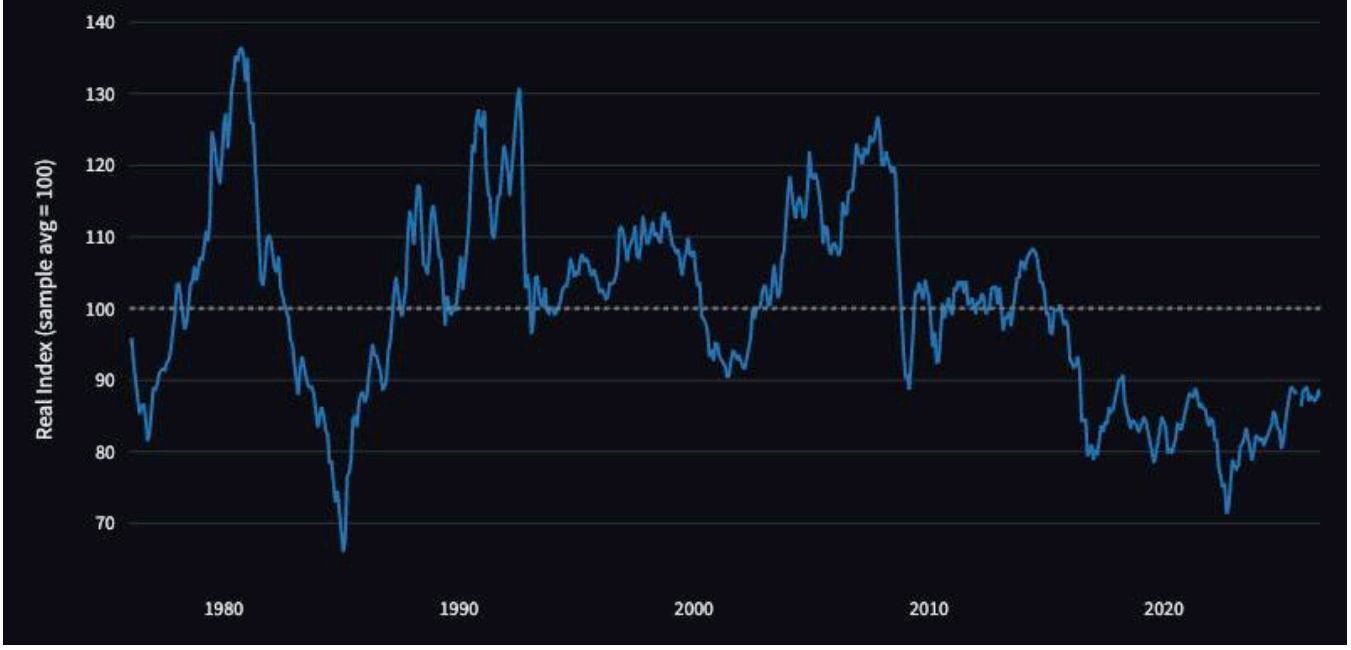
RER*	Valid months	Latest S(t)	Mean Real Index
1.4249	608 / 609	1.3411	100.00

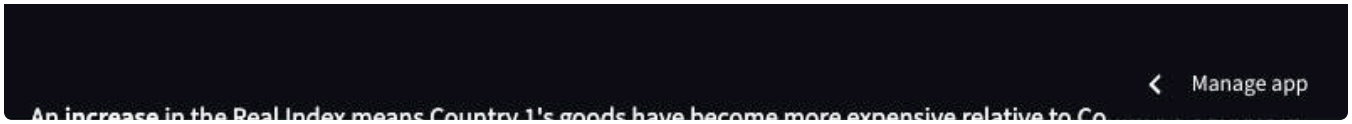
Months with a missing CPI or FX value are excluded from RER* and shown as gaps.

Chart

Real Nominal Both

United Kingdom (Country 1) vs United States (Country 2)





Nominal view. S(t) in actual USD per GBP, not normalized.

Share

Real Exchange Rate App

Monthly IMF CPI and exchange-rate data. S(t) = units of Country 2 currency per 1 unit of Country 1 currency; RER(t) = S(t) × P1(t) / P2(t); Real Index(t) = 100 × RER(t) / RER*.

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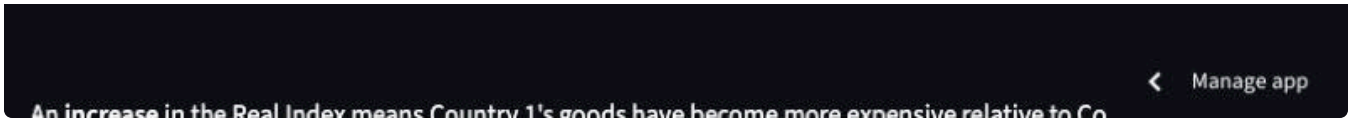
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Chart

Real Nominal Both

United Kingdom (Country 1) vs United States (Country 2)





Both view. Real Index on the left axis and S(t) on the right axis.

Share

Real Exchange Rate App

Monthly IMF CPI and exchange-rate data. S(t) = units of Country 2 currency per 1 unit of Country 1 currency; RER(t) = S(t) × P1(t) / P2(t); Real Index(t) = 100 × RER(t) / RER*.

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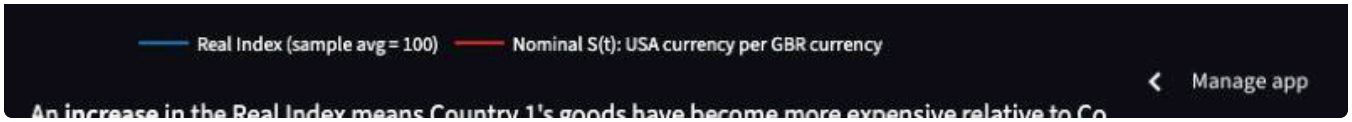
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Chart

Real | Nominal | Both

United Kingdom (Country 1) vs United States (Country 2)





Chat screenshots. Paste screenshots of the Claude chat here before submitting. The prompts they show are listed word for word in the Prompts section.

Checks (UK = Country 1, U.S. = Country 2)

Every check in step 5 of the assignment passed.

Check	Result
S is USD per GBP	Passed. The CSV column `S_USA_per_GBR` runs from about 1.25 to 1.64 over 2000–2025. On the live app the latest S(t) is 1.3411 (Sept 2026).
RER* changes with the sample dates	Passed. Full sample (Jan 1976–Sept 2026): RER* = 1.4249. Sample set to Jan 2000–Dec 2025: RER* = 1.377365.
Exported Real Index averages 100	Passed. Mean of `Real_Index` over the 311 valid months in the CSV = 100.0000.
Real, Nominal and Both charts	Passed. Both puts the Real Index on the left axis and S(t) in USD per GBP on the right axis.
Euro area works	Passed. The IMF publishes the Euro area (G163) index as HICP, not CPI, so the app uses HICP and says so on screen.

The IMF data has no U.S. CPI value for October 2025. The app marks that month `Valid_month = False`, leaves it out of RER*, and shows it as a gap in the chart.

Manual RER verification (June 2020)

The hand calculation gives RER = 1.152365 and Real Index = 83.66, the same as the app's exported row for 2020-06.

Inputs from row `2020-06` of `rer_GBR_USA_2000-01_2025-12.csv` (sample Jan 2000–Dec 2025):

Input	Value
FX1: GBP per USD	0.798597
FX2: USD per USD	1
P1: UK CPI	108.8
P2: U.S. CPI	118.225406
RER* for the sample	1.377365

Step 1, nominal rate in USD per GBP:

S = FX2 / FX1 = 1 / 0.798597 = 1.252195

Step 2, real exchange rate:

RER = S × P1 / P2 = 1.252195 × 108.8 / 118.225406 = 136.2388 / 118.2254 = 1.152365

Step 3, real index:

Real Index = 100 × RER / RER* = 100 × 1.152365 / 1.377365 = 83.66

Column in the CSV	App value	Hand calculation	Match
`S_USA_per_GBR`	1.252195	1.252195	Yes
`RER`	1.152365	1.152365	Yes
`Real_Index`	83.664492	83.66	Yes

In June 2020 the pound was about 16% weaker in real terms against the dollar than its 2000–2025 average.

Attachment: exported CSV

Submit rer_GBR_USA_2000-01_2025-12.csv with this document. It covers the sample Jan 2000–Dec 2025: 312 months, 311 of them valid.

Columns: `Month`, `FX1_GBR_per_USD`, `FX2_USA_per_USD`, `P1_CPI_GBR`, `P2_CPI_USA`, `S_USA_per_GBR`, `RER`, `RER_star`, `Real_Index`, `Valid_month`. The same file is in the GitHub repository.